

Grant LeRette-Kauffman

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Education

B.S. expected December 2026

University of Illinois Chicago

- GPA: 4.00/4.00
- Bachelor of Science in Computer Science

Skills

- **Languages:** C++, JavaScript, HTML, C, Python
- **Technologies:** CMake, OpenGL, PortAudio, FFTW, Git, Linux
- **Concepts:** Real-time media processing, multithreading, signal processing, data structures & algorithms, backend development

Projects

Real-Time Audio Visualizer

August 2025 - December 2025

C++ | OpenGL | PortAudio | FFTW | CMake

- Built cross-platform audio visualization tool processing audio files with real-time frequency spectrum analysis
- Designed thread-safe audio pipeline using PortAudio's ring buffer for lock-free data transfer between playback and analysis
- Implemented FFT-based audio analysis using FFTW library to process audio into 32 logarithmic frequency buckets
- Architected modular pipeline separating audio I/O, processing, and rendering components for maintainability and scalability

Interactive Maps Application

March 2025 - May 2025

Graph Algorithm Implementation | JSON Parsing

- Designed and implemented a custom graph data structure to represent geographical map data
- Parsed real-world map data from industry-standard formats (OSM) to create a navigable network
- Developed pathfinding algorithms (Dijkstra's) to determine optimal routes between locations
- Built a functional user interface allowing users to select locations and view mapped routes
- Optimized algorithm performance to handle large datasets with minimal processing time

Work Experience

March 2019 - July 2024

Dunkin Donuts, Batavia, IL - Part Time Barista and Shift Manager

- **Leadership & Team Coordination:** Managed team of 5-7 employees during shifts, created weekly schedules, and delegated tasks to maximize productivity and customer satisfaction
- **Problem Solving:** Resolved customer issues and POS system problems through rapid troubleshooting
- **Data & Inventory Management:** Tracked daily sales metrics, managed inventory levels, and coordinated with suppliers to ensure optimal stock levels